

TO YIN YU

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SUMMARY

Full-stack software engineer (B.S. CS, UMD) shipping applied AI products end-to-end — realtime voice AI on OpenAI's Realtime API over WebRTC, FastAPI backends with Postgres + Redis, and production deploys on Vercel, Render, Neon, and Upstash. Open to SWE roles.

TECHNICAL SKILLS

Languages: TypeScript, Python, JavaScript, SQL

Web & Backend: Next.js, React, FastAPI, Tailwind CSS, PostgreSQL, Redis

Applied AI & Realtime: OpenAI Realtime API, WebRTC, ElevenLabs, LLM tool use, TensorFlow, Keras

Infra & Deploys: Docker, Git, Linux/Unix, Vercel, Render, Neon, Upstash, Netlify

PROJECT EXPERIENCE · Independent Software Projects, 2024 – Present

SoloMock — AI Voice Mock Interviewer | Next.js, TypeScript, OpenAI Realtime API, WebRTC, Monaco, Tailwind CSS, Vercel

solomock.com · 2025

- Built a realtime voice mock-interview app on OpenAI's Realtime API over WebRTC: the candidate speaks through coding problems while the AI probes complexity, gives Socratic hints, and reads code via debounced Monaco-editor snapshots streamed over a WebRTC data channel.
- Authored 15 structured per-problem "interviewer briefs" (solution tree, 4-rung hint ladder, follow-ups, and edge cases) that drive interviewer behavior through the system prompt.
- Hardened sessions against cost and abuse: server-minted ephemeral API keys so the browser never holds the secret, Upstash sliding-window rate limits, a 15-minute session cap, and a Discord-webhook request flow with manual allowlisting.

Reachspan — AI Lead-Gen via Social Listening | Python, FastAPI, Claude / LLM Tool Use, Vercel Serverless, Cal.com API, Tailwind CSS

reachspan.ai · 2026

- Co-founded Reachspan AI LLC and built its AI lead-gen engine: a Python + FastAPI + Claude engine that monitors Reddit and other channels, classifies buying intent against a configurable ruleset, and drafts brand-safe replies queued for human review.
- Built the marketing site with Vercel serverless functions wired to the Cal.com API so prospects book audit calls end-to-end with auto-confirmation.
- Designed the classification and human-in-the-loop pipeline (YAML-defined criteria, scored leads persisted to SQLite) to keep automated outreach on-brand.

all-in-one-URL — Short URLs, QR Codes & Barcodes | Python, FastAPI, PostgreSQL, Redis, Docker, React, TypeScript, Tailwind CSS

all-in-one-url.vercel.app · 2024

- Built a FastAPI backend for short URLs, QR codes, and barcodes with per-resource click/scan analytics, Redis-backed caching and counter batching, and tiered SlowAPI rate limiting.
- Implemented JWT auth with bcrypt hashing and per-resource ownership (preserving anonymous resources); containerized the two-tier app with Docker Compose and deployed across Vercel, Render, Neon (Postgres), and Upstash (Redis).

Gasolytics — US Gas Price Map | Next.js 16, React 19, TypeScript, d3-geo, Vercel Blob, Vercel

gasolytics.com · 2026

- Built an interactive US gas-price map with a d3-geo choropleth projected to SVG server-side (so no d3 ships to the client), metro price pins, wheel-zoom/drag-pan, and per-state detail panels.
- Expanded it into a national fuel-economics toolkit — 51 state and metro pages, a trip-cost calculator, an EV-vs-gas suite, and gas-tax / border-gap / affordability analyses — all fed by daily AAA snapshots from a separate data pipeline.

Throughline — Technical Learning Ecosystem | Docusaurus, Astro, React, TypeScript, MDX

hub · web-dev · ai · security · 2026

- Built Throughline, a connected source-available learning ecosystem: six first-principles guides (web dev, AI, security, cloud, data engineering, programming basics) on one shared design system, an Astro hub, and Shipyard — a build-a-project-and-AI-grade-it platform (Claude-as-judge).

EDUCATION

University of Maryland, College Park — College Park, MD

Bachelor of Science, Computer Science · December 2022

Relevant coursework: Data Structures, Artificial Intelligence, Data Science, Computer Systems, Web Development.